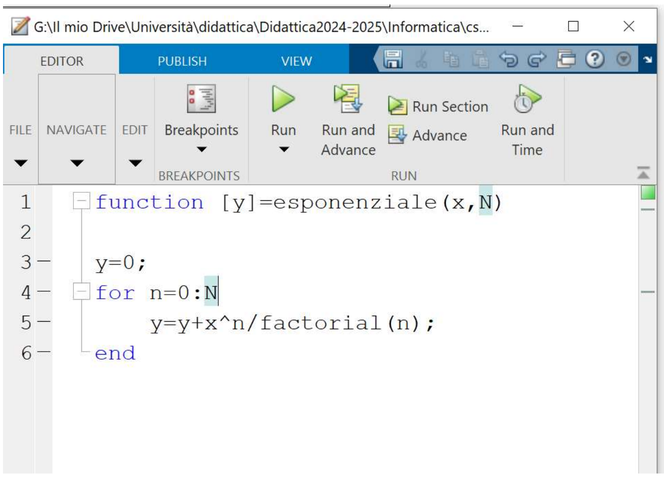
3/3/2025

lunedì 3 marzo 2025 15:48

for $n=0:N$
    istruzioni
end

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^N}{N!} = \sum_{n=0}^{N} \frac{x^n}{n!}$$



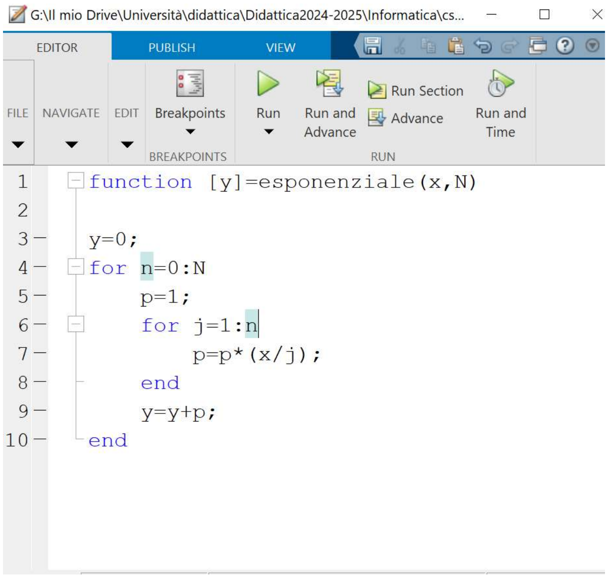

```
function [y]=esponenziale(x,N)

y=0;
for n=0:N
    y=y+x^n/factorial(n);
end
```

per x ed n grandi $\rightarrow$ x^n overflow

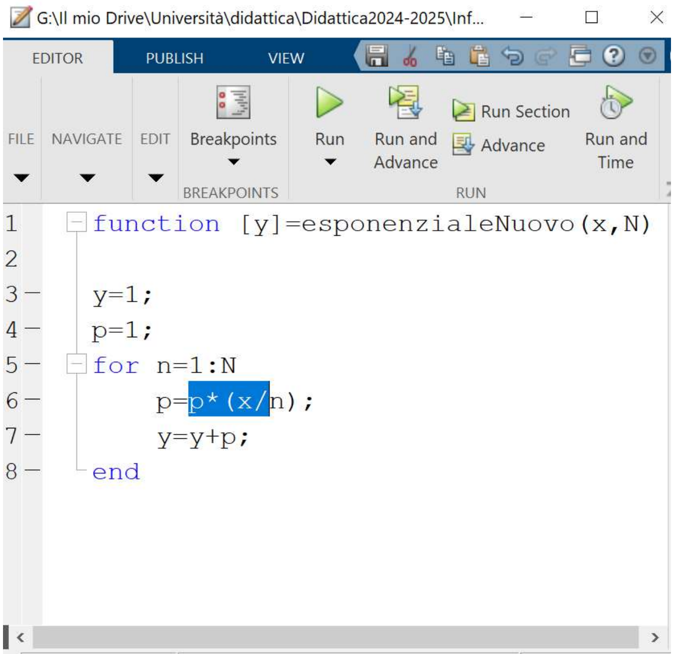
$n! = n(n-1)(n-2)\ldots 1$ overflow

$$\frac{x^n}{n!} = \frac{x \cdot x \cdot x \cdot \ldots \cdot x}{N\cdot(N-1)(N-2)\cdot\ldots\cdot 1} = \left(\frac{x}{n}\right)\cdot\left(\frac{x}{n-1}\right)\cdot\left(\frac{x}{n-2}\right)\cdot\ldots\cdot\left(\frac{x}{1}\right)$$



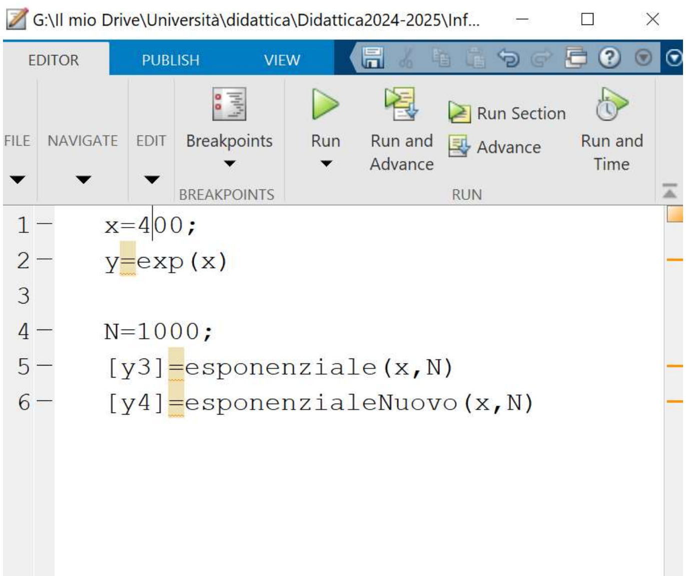
```
function [y]=esponenziale(x,N)

y=0;
for n=0:N
    p=1;
    for j=1:n
        p=p*(x/j);
    end
    y=y+p;
end
```



```
function [y]=esponenzialeNuovo(x,N)

y=1;
p=1;
for n=1:N
    p=p*(x/n);
    y=y+p;
end
```



```
x=400;
y=exp(x)

N=1000;
[y3]=esponenziale(x,N)
[y4]=esponenzialeNuovo(x,N)
```

Command Window

New to MATLAB? See resources for Getting Started.

```
    5.221469689764124e+173

>> esercizio

y =

    5.221469689764144e+173


y3 =

    5.221469689764138e+173


y4 =

    5.221469689764138e+173
```